Sri Eshwar College of Engineering
Coimbatore | Tamilnadu
An Autonomous Institution
Affiliated to Anna University, Chennai

Department of Computer Science and Engineering

Course name /Course code: R19CS481/ Project Phase 1

AssistiveVision : YOLO driven multilingual object detection with angle awareness for mobile devices

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OVERVIEW

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CHALLENGE STATEMENT

- Most mobile-based object detection systems only recognize objects without indicating their spatial location relative to the user.
- Apps for the visually impaired, like Microsoft Seeing AI and Google Lookout, lack directional guidance.
- Multilingual support is limited, often only covering English and a few major languages.
- These limitations hinder situational awareness, usability, and accessibility for millions globally.

LITERATURE SURVEY / BACKGROUND STUDY

S. No	TITLE	AUTHOR	NAME OF ORGANIZATION &YEAR	DESCRIPTION
1.	You Only Look Once: Unified, Real-Time Object Detection	Joseph Redmon, Santosh Divvala, Ross Girshick, Ali Farhadi	IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016	• Introduced YOLO, a real-time object detection system performing detection in a single pass. • Significantly faster than region-based methods like R-CNN.

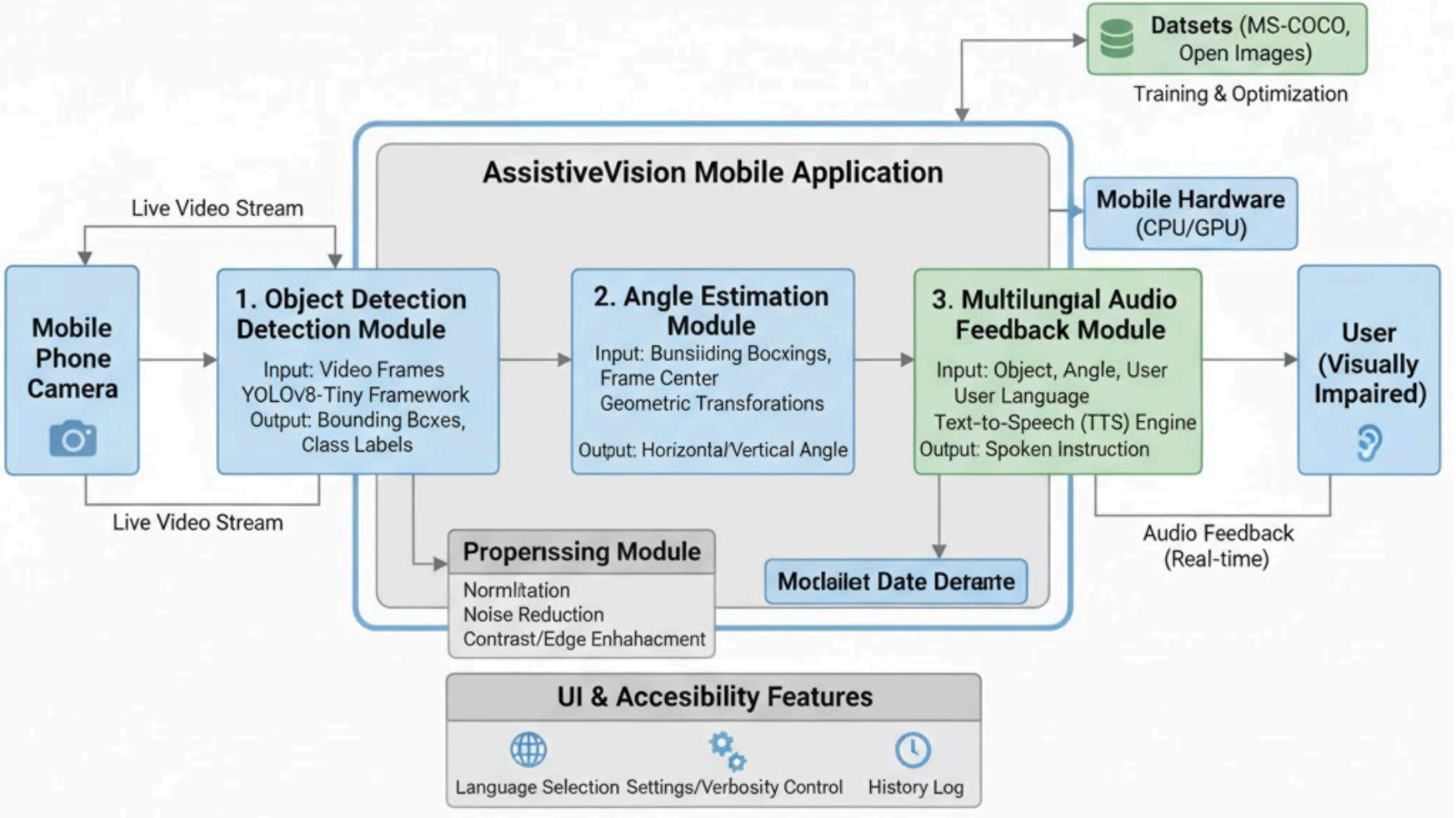
EXISTING METHOD

1. Earlier object detection methods like Haar Cascades and HOG provided low accuracy and were not suitable for complex or real-time scenarios.
2. R-CNN, Fast R-CNN, and Faster R-CNN improved detection accuracy but required high computational power, making them unsuitable for mobile devices.
3. YOLO and SSD introduced single-shot detection, enabling real-time object detection with better speed and efficiency.
4. Existing assistive apps like Microsoft Seeing AI and Google Lookout perform recognition but lack spatial awareness and multilingual audio feedback.

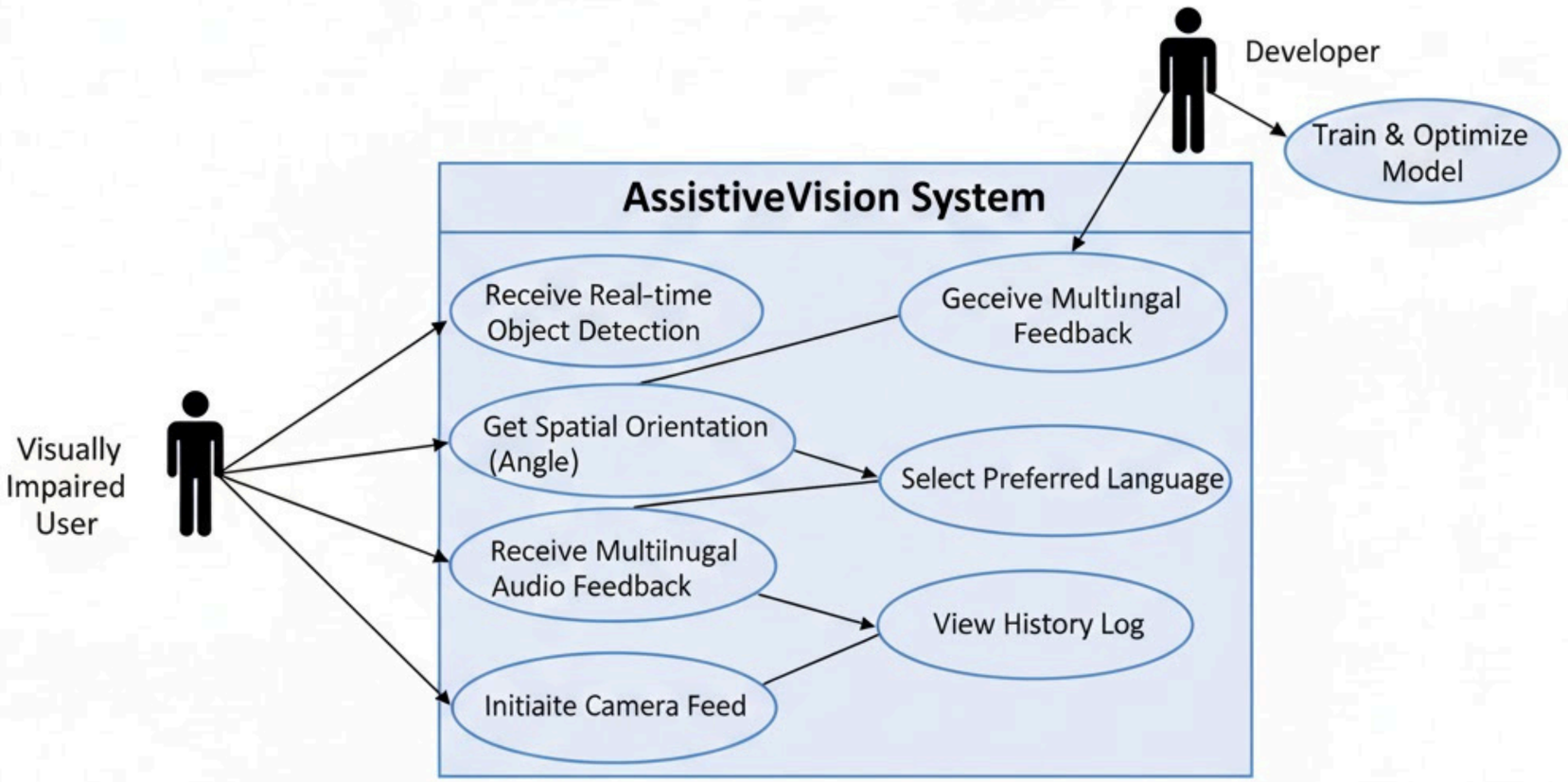
PROPOSED / INNOVATIVE METHOD

1. YOLOv8-Tiny Model – This tool helps in spotting objects quickly and accurately, perfect for use on mobile devices.
2. Angle Estimation Module – It figures out where each object is in relation to you, whether it's to your left, right, or straight ahead.
3. Multilingual Audio Feedback – This feature turns what the app detects into spoken words, using various languages through Text-to-Speech (TTS) technology.
4. Preprocessing Techniques – It involves steps like normalizing, reducing noise, and enhancing contrast to make detection more precise.
5. Motion Tracking – It employs Optical Flow and Kalman Filter to keep track of moving objects smoothly as they appear in different frames.

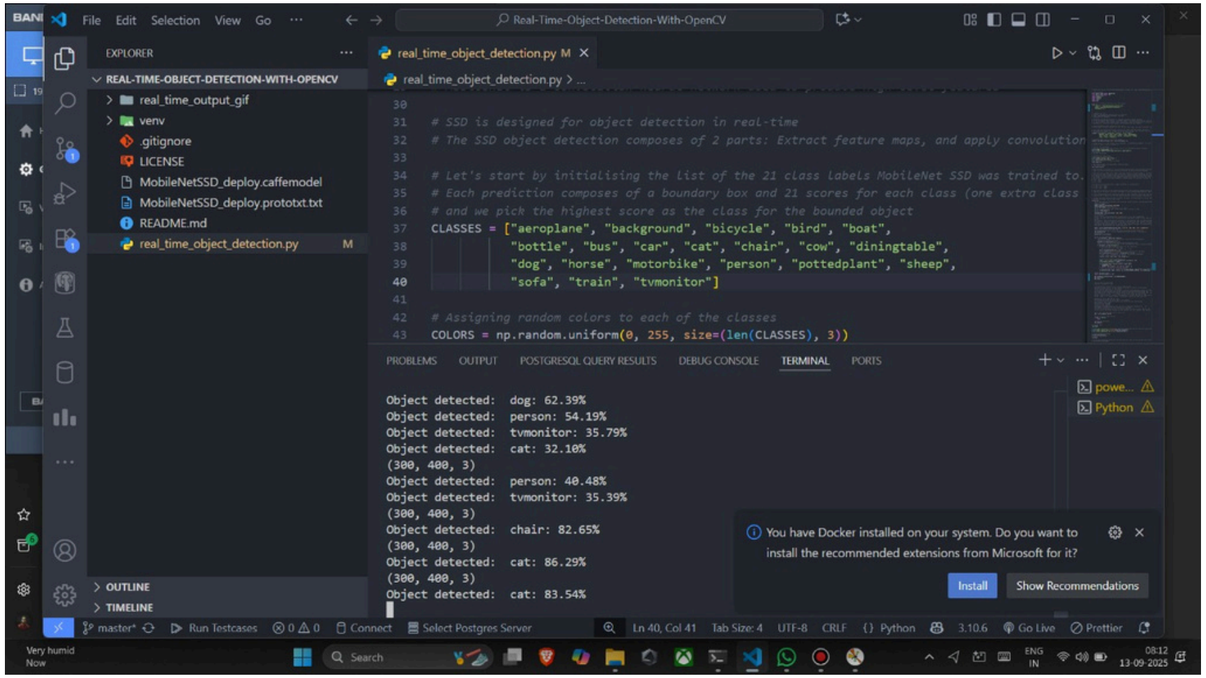
AssistiveVision: System Architecture Block Diagram

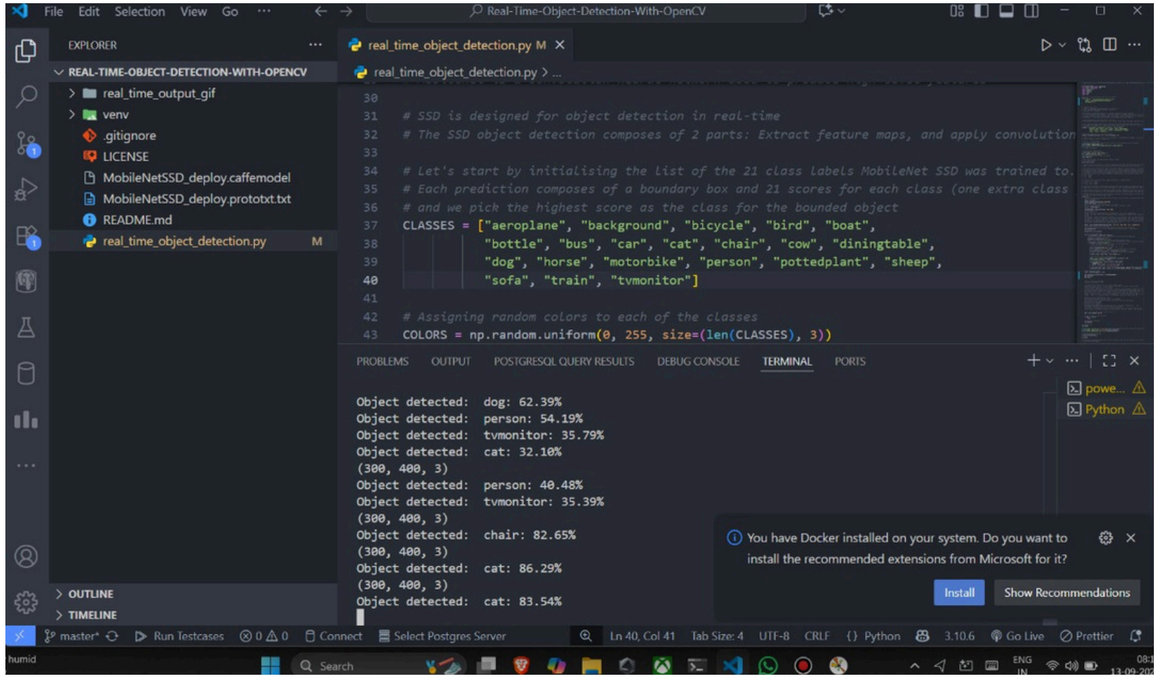


AssistiveVision: Use Case Diagram



SOFTWARE/HARDWARE IMPLEMENTATION IMAGE

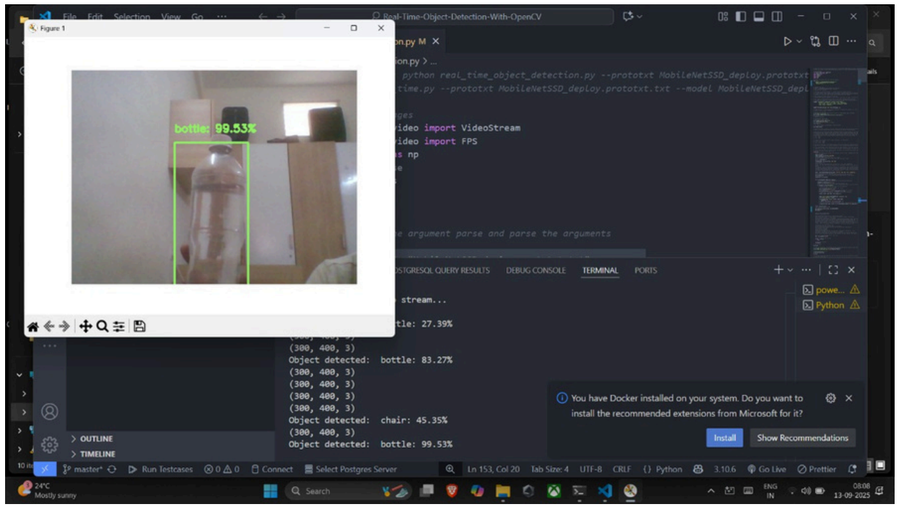


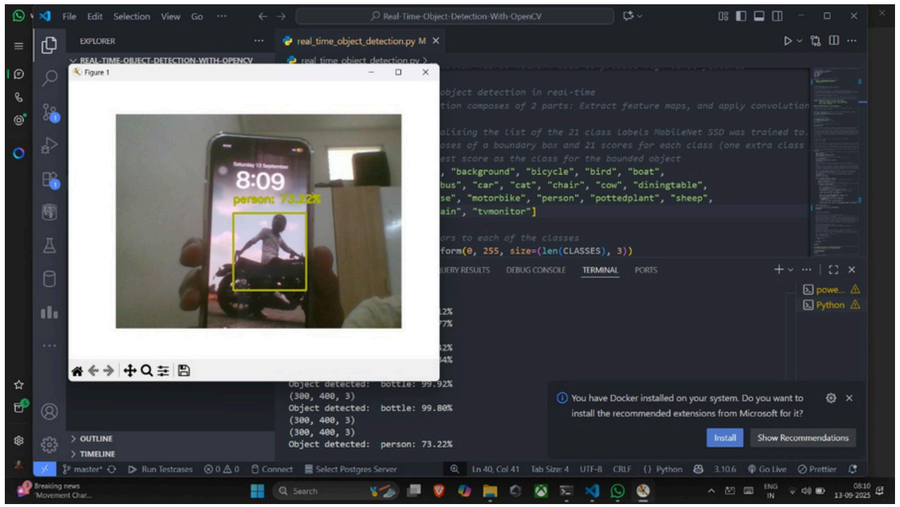


DESCRIPTION OF SOFTWARE/HARDWARE IMPLEMENTATION

S.No	SOFTWARE SPECIFICATIONS
1.	Flutter
2.	Python
3.	VSCode
4.	Kaggle

EXPERIMENTAL RESULTS





CONCLUSION & FUTURE WORK

The AssistiveVision application demonstrates a novel approach to mobile object detection by integrating YOLO-based real-time detection with crucial accessibility features. The system successfully bridges the gap between mere object recognition and actionable situational awareness for visually impaired users.

REFERENCES

[1] A. Gulshan et al., “Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs,” JAMA, vol. 316, no. 22, pp. 2402–2410, 2016.

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THANK YOU